aszad-akib/hg-on-gold-mlip · Public repository

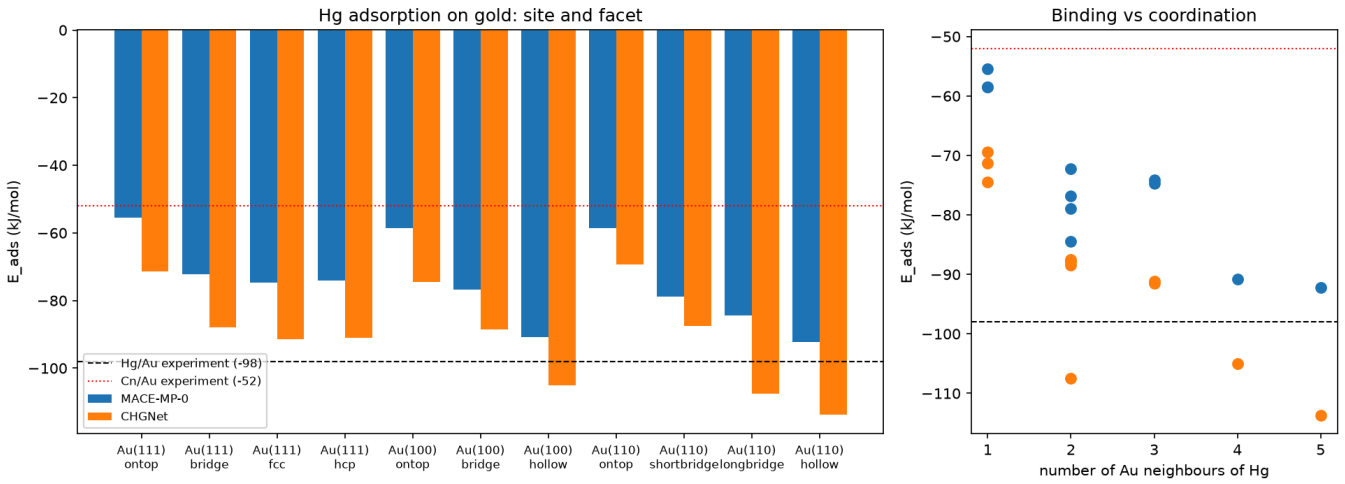
https://github.com/aszad-akib/hg-on-gold-mlip

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Hg adsorption on gold surfaces with machine-learned interatomic potentials (ASE)

Mercury is the lighter homolog of **copernicium (Cn, Z = 112)**, whose chemistry is studied by single-atom thermochromatography on gold-covered detectors. This repository extends the baseline ASE scripts from `miroi/open-collection` into a production workflow. It computes Hg adsorption energies on **all 11 high-symmetry sites of Au(111), Au(100) and Au(110)** with two universal MLIPs, **MACE-MP-0** and **CHGNet**, and checks convergence.

Everything was run from a Linux terminal (WSL2 Ubuntu) on a laptop CPU. Each script finishes in about 5 minutes.



Key findings

1. **Site preference is robust.** Both models rank the sites ontop < bridge < hollow on every facet, and binding grows with the number of Au neighbours (right panel above).
2. **Open facets bind more strongly.** Au(100) and Au(110) bind Hg 14–22 kJ/mol more strongly than the close-packed Au(111). A rough or polycrystalline detector will hold Hg more strongly than an ideal (111) surface.
3. **The potential matters as much as the facet.** The two MLIPs differ by 9–23 kJ/mol. CHGNet (−91.5 to −113.7 kJ/mol at the best sites) brackets the experimental value of about −98 kJ/mol, while MACE-MP-0 underbinds every site (−74.7 to −92.2).
4. **Hg binds much more strongly than Cn** (52 kJ/mol) in both models, as in experiment.
5. **The baseline set-up underbinds by about 5 kJ/mol.** The demo scripts build the slab at a = 4.08 Å, but each MLIP has its own lattice constant (MACE 4.176 Å, CHGNet 4.169 Å). That puts the slab under about 2 % strain. The MACE demo also tests only the bridge site.
6. **The default Lennard-Jones demo gives −0.01 eV,** because ASE's `LennardJones()` defaults to σ = 1 Å and ε = 1 eV for every pair. A pair potential must be parametrised for Hg–Au before it means anything.

Results (kJ/mol)

Facet	Site	Au neighbours	MACE-MP-0	CHGNet
Au(111)	ontop	1	−55.4	−71.3
	bridge	2	−72.2	−87.9
	fcc	3	**−74.7**	**−91.5**
	hcp	3	−74.1	−91.1
Au(100)	ontop	1	−58.5	−74.5
	bridge	2	−76.8	−88.5
	hollow	4	**−90.8**	**−105.0**
Au(110)	ontop	1	−58.5	−69.4
	short bridge	2	−78.9	−87.5
	long bridge	2	−84.5	−107.4
	hollow	5	**−92.2**	**−113.7**

Facet	Site	Au neighbours	MACE-MP-0	CHGNet
Experiment (Hg/Au)			≈ −98	

Convergence: Au(111) fcc site (kJ/mol)

Test	Values	MACE-MP-0	CHGNet
Layers	3 / 4 / 5 / 6	−74.5 / −74.7 / −74.8 / −74.7	−91.0 / −91.5 / −91.1 / −91.0
Cell	2×2 / 3×3 / 4×4	−73.6 / −74.7 / −75.1	−85.9 / −91.5 / −92.4
Vacuum (Å)	8 / 12 / 16	−74.7 / −74.7 / −74.7	−91.5 / −91.5 / −91.4
Lattice (Å)	own / 4.08	−74.7 / −69.9	−91.5 / −86.5

Slab thickness and vacuum change the result by less than 0.5 kJ/mol. Going from 3×3 to 4×4 changes it by less than 1 kJ/mol, so the 3×3 production cell is converged.

Method

E_ads = E(slab + Hg) − E(slab) − E(Hg atom), with all three terms from the same potential (1 eV = 96.485 kJ/mol).

- The lattice constant comes from each model's own equation-of-state fit.
- Slabs are 3×3 with 4 layers (111, 100) or 6 layers (110) and 12 Å of vacuum. The bottom half is fixed, and the rest is relaxed with BFGS to f_max = 0.02 eV/Å.
- Hg is relaxed with its x, y fixed on each site (`FixCartesian`), so each energy belongs to its labelled site.
- The isolated Hg atom is computed in an 18 Å box. CHGNet's "isolated atom" warning does not affect the result: the energy equals that of Hg held 10 Å above the slab.

Repository layout

```
production.py     all sites on Au(111)/(100)/(110)   ->  results_<model>.csv
convergence.py    layers, cell, vacuum, lattice tests ->  conv_<model>.csv
hg_sites.py       first modification: 4 sites on Au(111)
plot.py           makes figures/Eads_facets.png
results/          CSV tables
structures/       relaxed geometries (.xyz, open with `ase gui`)
report/main.tex   full write-up (LaTeX)
```

Reproduce

```
python -m venv ase-env && source ase-env/bin/activate
pip install -r requirements.txt
nohup python production.py mace   > prod_mace.log   2>&1 &
nohup python production.py chgnet > prod_chgnet.log 2>&1 &
python convergence.py mace; python convergence.py chgnet
mkdir -p results structures figures
mv results_*.csv conv_*.csv results/; mv *.xyz structures/
python plot.py
```

Limitations and next steps

- MACE-MP-0 and CHGNet are trained on PBE data with few adsorbate configurations and no dispersion correction. CHGNet's agreement with experiment may be partly fortuitous.
- Spin–orbit coupling is absent, and it is decisive for Cn. These potentials cannot be transferred to Cn without relativistic training data.
- E_ads is a static 0 K energy. Experiment measures an enthalpy on a non-ideal surface.
- Next: D3 dispersion, newer foundation models (MACE-MPA-0, MACE-OMAT-0, SevenNet), stepped and reconstructed surfaces, and fine-tuning on relativistic DFT reference points.

References

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- R. Eichler, "First foot prints of chemistry on the shore of the Island of Superheavy Elements", *J. Phys.: Conf. Ser.* **420**, 012003 (2013), arXiv:1212.4292.
- A. H. Larsen *et al.*, "The atomic simulation environment", *J. Phys.: Condens. Matter* **29**, 273002 (2017).
- I. Batatia *et al.*, "A foundation model for atomistic materials chemistry" (MACE-MP-0), arXiv:2401.00096.
- B. Deng *et al.*, "CHGNet as a pretrained universal neural network potential", *Nat. Mach. Intell.* **5**, 1031 (2023).
- Baseline scripts: M. Iliaš, `miroi/open-collection`.

Based on project 411 proposed by Prof. M. Iliaš for the INTEREST programme (JINR Dubna).